

Disproof of conjecture 00000002308

earthking11

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Verdict: FALSE

We disprove conjecture 00000002308 as stated. The refutation is unconditional and completely elementary. Under the standard meaning of *complete* — trivial center and every automorphism inner, equivalently $G \cong \text{Aut}(G)$ via conjugation — the symmetric group S_3 , of order 6, is a solvable complete group. Every group of order 2, 3, 4, 5 is abelian, hence has non-trivial center and is not complete. Therefore the smallest non-trivial solvable complete group has order 6, not the claimed $2^{11} \cdot 3 = 6144$.

1 The conjecture, quoted verbatim

From `conjectures/00000002308.md`:

English. Definition: A complete group satisfies $G = \text{Aut}(G)$. Conjecture: The order of the smallest nontrivial solvable complete group is $2^{11} \cdot 3$ (an explicit discovery); its structure is a minimal example of a semidirect product of a 2-group.

The conjecture is filed in both English and Chinese; the Chinese original is reproduced verbatim (in Unicode) in `README.md`. The two versions are identical in content. The PDF font does not embed CJK glyphs, so the Chinese is kept in the markdown copy rather than duplicated here.

2 The definition

Definition 1 (Complete group). A group G is *complete* if

1. its center is trivial, $Z(G) = \{1\}$, and
2. every automorphism of G is inner, $\text{Aut}(G) = \text{Inn}(G)$.

Equivalently, the conjugation map $G \rightarrow \text{Aut}(G)$, $g \mapsto (x \mapsto gxg^{-1})$, is an isomorphism. The trivial group is excluded as “nontrivial”.

For a centerless group the conjugation map is injective, so $\text{Inn}(G) \cong G$; completeness is then exactly $\text{Aut}(G) = \text{Inn}(G)$, which gives $|G| = |\text{Aut}(G)|$ and $G \cong \text{Aut}(G)$. Both formulations are equivalent.

3 The witness S_3

We model S_3 as the six permutations of $\{0, 1, 2\}$. In cycle notation

$$S_3 = \{ 1, (12), (01), (012), (021), (02) \}.$$

The transposition $\tau = (01)$ and the 3-cycle $\sigma = (012)$ generate S_3 ; the six words

$$1, \quad \tau\sigma, \quad \tau, \quad \sigma, \quad \sigma^2, \quad \tau\sigma^2$$

are pairwise distinct and exhaust the group. (Both facts are checked by a finite computation, in `reproduce.py` and in the Lean certificate `lean4/Main.lean`.)

Theorem 2. S_3 is complete.

Proof. Trivial center. A direct computation over the six elements shows that the only element commuting with all of S_3 is the identity:

$$Z(S_3) = \{z \in S_3 : \forall a \in S_3, za = az\} = \{1\}.$$

Indeed the three transpositions $(12), (01), (02)$ are not central and neither are the two 3-cycles: for instance $(01)(012) = (021) \neq (012)(01) = (12)$.

Every automorphism is inner. The group S_3 has exactly six automorphisms. This is verified by enumerating all $6! = 720$ bijections $f : S_3 \rightarrow S_3$ and retaining those satisfying $f(ab) = f(a)f(b)$ for all a, b . Exactly six of the 720 bijections are homomorphisms, so

$$|\text{Aut}(S_3)| = 6.$$

The same enumeration shows the six conjugations $\text{Inn}(S_3) = \{x \mapsto gxg^{-1} : g \in S_3\}$ are pairwise distinct, so $|\text{Inn}(S_3)| = 6$; each is a bijective homomorphism, and the two sets have the same six elements:

$$\text{Aut}(S_3) = \text{Inn}(S_3).$$

Hence S_3 is centerless and all its automorphisms are inner, i.e. it is complete. Consequently $S_3 \cong \text{Aut}(S_3)$ and $|\text{Aut}(S_3)| = |\text{Inn}(S_3)| = 6 = |S_3|$. $\square$

Equivalent generator-based reduction (also carried out in the Lean certificate, which uses it for speed): since τ, σ generate, a homomorphism is determined by the pair $(u, v) = (f(\tau), f(\sigma))$, and the six words above force f to be

$$1 \mapsto 1, \quad \tau\sigma \mapsto uv, \quad \tau \mapsto u, \quad \sigma \mapsto v, \quad \sigma^2 \mapsto v^2, \quad \tau\sigma^2 \mapsto uv^2.$$

Enumerating the $6 \times 6 = 36$ pairs (u, v) , exactly 10 extend to endomorphisms (1 trivial, 3 with image of order 2, and 6 automorphisms), again giving $|\text{Aut}(S_3)| = 6$ with the automorphisms exactly the six conjugations.

4 Solvability and the semidirect-product structure

Theorem 3. S_3 is solvable and is a semidirect product of a 2-group,

$$S_3 = A_3 \rtimes \langle \tau \rangle \cong \mathbb{Z}/3 \rtimes \mathbb{Z}/2.$$

Proof. Let $A_3 = \{1, \sigma, \sigma^2\}$, the alternating group, of order 3. Then A_3 is abelian (it is cyclic of order 3, generated by σ with $\sigma^3 = 1$), it is normal in S_3 (being the kernel of the sign homomorphism, or directly $gag^{-1} \in A_3$ for all g, a), and it has index 2, since $|S_3|/|A_3| = 6/3 = 2$. The quotient $S_3/A_3 \cong \mathbb{Z}/2$ is abelian. Thus the subnormal series

$$1 \triangleleft A_3 \triangleleft S_3$$

has abelian factors $A_3 \cong \mathbb{Z}/3$ and $S_3/A_3 \cong \mathbb{Z}/2$; hence S_3 is solvable.

For the semidirect product let $T = \langle \tau \rangle = \{1, \tau\}$, a 2-group of order $2 = 2^1$. Then T is a subgroup of order 2, $A_3 \cap T = \{1\}$ (their only common element is the identity), and the product set

$$A_3T = \{at : a \in A_3, t \in T\}$$

has $3 \cdot 2 = 6 = |S_3|$ elements, hence equals S_3 . With A_3 normal, this is an internal semidirect product $S_3 = A_3 \rtimes T$. The action of T on A_3 by conjugation is non-trivial — for example $\tau\sigma\tau^{-1} = \sigma^{-1} \neq \sigma$ — so the product is not the direct product $\mathbb{Z}/6$, and $S_3 \cong \mathbb{Z}/3 \rtimes \mathbb{Z}/2$ with inversion action. $\square$

This is *exactly* the structural description in the conjecture — “a semidirect product of a 2-group” — but realised at order 6, not at $2^{11} \cdot 3$.

5 Minimality: no group of order 2, 3, 4, 5 is complete

Lemma 4. *Every group of order 2, 3, 4 or 5 is abelian.*

Proof. For prime order (2, 3, 5) the statement is Lagrange’s theorem: a non-identity element has order dividing the prime, hence equal to it, so the element generates the whole group, which is therefore cyclic and abelian. For order 4: if G has an element g of order 4 then $G = \langle g \rangle$ is cyclic and abelian; otherwise every non-identity element has order 2, so $g^2 = 1$ for all g and G is abelian. (This case analysis is exhaustive because the order of an element divides $|G| = 4$.)

An independent, exhaustive machine check is carried out in `reproduce.py`: every group of order n can be relabelled so that its multiplication table is a *reduced Latin square* (first row and column equal to $0, 1, \dots, n-1$). Enumerating all reduced Latin squares for $n = 2, 3, 4, 5$ and retaining the associative ones yields 1, 1, 4, 6 tables respectively, and *all* of them are abelian. Thus for $n \in \{2, 3, 4, 5\}$ every group is abelian. $\square$

Corollary 5. *No group of order 2, 3, 4 or 5 is complete, and 6 is the smallest order of a non-trivial solvable complete group.*

Proof. A group of order 2, 3, 4 or 5 is abelian (Lemma 4), so its center is the whole group, which is non-trivial (the groups are non-trivial). By Definition 1 such a group is not complete. The group S_3 of order 6 is solvable (Theorem 3) and complete (Theorem 2), so the least order of a non-trivial solvable complete group is exactly 6. $\square$

6 The claimed order is wrong

Corollary 6 (Refutation). *The assertion “the order of the smallest nontrivial solvable complete group is $2^{11} \cdot 3$ ” is false. In fact*

$$6 < 2^{11} \cdot 3 = 6144,$$

and 6 is the true minimal order.

Proof. By Corollary 5 the minimal order is 6, and $2^{11} \cdot 3 = 2048 \cdot 3 = 6144 > 6$. □

The claim is wrong under *every* reading, not just the solvable one:

- **Solvable reading** (as filed). The smallest non-trivial solvable complete group is S_3 of order 6 (Corollary 5), and $6 < 6144$.
- **Non-solvable reading**. A folklore theorem states that the symmetric group S_n is complete for every $n \neq 2, 6$ (and S_6 is exceptional because $\text{Aut}(S_6)/\text{Inn}(S_6) \cong \mathbb{Z}/2$). Hence S_3 is complete in this sense too. Even the smallest *non-solvable* complete group is S_5 , of order 120: the alternating group A_5 of order 60 is not complete, since $\text{Out}(A_5) = \mathbb{Z}/2 \neq 1$. Again $120 < 6144$.
- **Order of a semidirect product of a 2-group**. The witness $S_3 = \mathbb{Z}/3 \rtimes \mathbb{Z}/2$ is the smallest nontrivial semidirect product of cyclic groups of coprime prime order, at order 6.

The number $2^{11} \cdot 3 = 6144$ is not the order of the smallest example under any of these readings.

7 Formalisation

The refutation is formalised in the Lean 4 project under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`). The model is S_3 on `Fin 6` with explicit multiplication and inverse tables. The main statements are

```
theorem card_S3 : (List.finRange 6).length = 6
theorem center_trivial : (the only central element is the identity)
theorem card_inn : InnList.length = 6
theorem card_aut : (candidates.filter isAut).length = 6
theorem aut_eq_inn : (every automorphism is inner)
theorem A3_abelian / A3_normal / A3_index2
theorem solvable_S3
theorem six_lt_6144 : 6 < 2 ^ 11 * 3
theorem conjecture_00000002308_false
```

The certificate uses the generator-based reduction: it verifies by a finite `decide` check that the two generators τ, σ have closure all six elements, then enumerates the 36 pairs of generator images $(f(\tau), f(\sigma))$ and counts the resulting automorphisms. This gives $|\text{Aut}(S_3)| = 6$ in a ‘lake build’ of about 2.6 seconds, rather than the ≈ 4 m 45 s that a direct enumeration of all $6^6 = 46656$ functions would take. The axiom audit (`Check.lean`, via `#print axioms`) reports no `sorryAx`; most theorems depend only on `propext`, and `card_S3` and `six_lt_6144` on no axioms at all. See `lean4/README.md` for the statement table, the encoding pitfalls, and the measured build times.

8 Reproducibility

```
$ python3 reproduce.py
... (group axioms, center, all 720 bijections, Aut = Inn, solvability, semidirect
product, exhaustive small-order check, order comparison)
PASS: all checks verified; conjecture 00000002308 is FALSE.
```

The script uses the Python 3 standard library only. It builds S_3 as the permutations of $\{0, 1, 2\}$, computes the center, enumerates all 720 bijections to obtain $|\text{Aut}(S_3)| = 6$, shows $\text{Aut}(S_3) = \text{Inn}(S_3)$ with $|\text{Inn}(S_3)| = 6$, verifies solvability and the decomposition $S_3 = \mathbb{Z}/3 \rtimes \mathbb{Z}/2$, exhaustively checks that every group of order 2, 3, 4, 5 is abelian, asserts $6 < 6144$, prints PASS and exits 0 exactly when every check holds.

Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic main.tex`.
- **PDF** — at `build/main.pdf`, produced by `tectonic --outdir build main.tex`.
- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (library `Main`, no dependencies) and `name = "tlmc2308"`, `Main.lean`, `Check.lean` (`#print axioms`), and `lean4/README.md`. `lake build` exits 0 in about 2.6 seconds and the audit reports no sorryAx.